

CSSE 220 – Object-Oriented Software Development
Rose-Hulman Institute of Technology

Worksheet 19

Name (Print): _____ Section: _____

1. Summary Table for MergeSort:

Case	BigO
Worst	
Best	

2. Suppose the merge sort algorithm from class is applied to the initial array.

9	5	7	10	18	1	12	8	16	4
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Show the state of the two sub-arrays immediately before the final merge.

--	--	--	--	--	--	--	--	--	--

3. Provide the order of the **five** time complexity families as input size increases:

4. Identify the Big-O

```

1  public static long method1(int[] arr) {
2      long result = 0;
3      int n = arr.length;
4      for (int j = 0; j < n; j++) {
5          for (int k = 0; k < 10000; k++) {
6              result += (j * k);
7          }
8      }
9      for (int j = 0; j < n; j++) {
10         result += (2 * j) - 3;
11     }
12     return result;
13     // Big O -----
14 }
15

```

5. Tracing

```

1 class A {
2     void m() { System.out.println("A.m"); }
3     void n() { System.out.println("A.n"); }
4 }
5 class B extends A {
6     @Override void m() { System.out.println("B.m"); }
7 }
8 class C extends B {
9     @Override void m() { System.out.println("C.m"); }
10 }
11 class D extends A {
12     @Override void n() { System.out.println("D.n"); }
13 }
14 A v1 = new A();
15 B v2 = new B();
16 A v3 = new B();
17 A v4 = new C();

```

v1.m(); _____

v2.m(); _____

v3.m(); _____

v4.m(); _____

```

1 class A {
2     void m() { System.out.println("A.m"); }
3 }
4 class B extends A {
5     void onlyInB() { System.out.println("B.onlyInB"); }
6 }
7 class C extends B {
8     void cOnly() { System.out.println("C.cOnly"); }
9 }
10 class D extends A { }
11 A v1 = new C();
12 A v2 = new B();
13 B v3 = new B();

```

((B) v1).onlyInB(); _____

((C) v1).cOnly(); _____

((C) v2).cOnly(); _____

A a3 = v3; _____